1. **Aim of the practical:** Create a simple client server communication application while creating a HTML with form and two input fields like First Name and Last Name. Submit the data to a servlet where the form data is retrieved and displayed. Create form with GET method and another with POST method.

2. **Objective:** To implement a servlet-based registration system that handles both GET and POST requests, displaying user registration results.

3. **Commands/Steps:**

CODE: index.html

```
<!DOCTYPE html>
<!--
Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Html.html to edit this template
-->
<html>
<head>
<title>TODO supply a title</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```html
<link rel="stylesheet" href="index.css">
</head>
<body>
<div class="registration-container">
<h1>Registration Form</h1>
<form action="register" method="post">
<div class="form-group">
<label for="name">Name:</label>
<input type="text" id="name" name="name" required>
</div>
<div class="form-group">
<label for="email">Email:</label>
<input type="email" id="email" name="email" required>
</div>
<div class="form-group">
<label for="password">Password:</label>
<input type="password" id="password" name="password" required>
</div>
<div class="form-group">
<label for="confirm-password">Confirm Password:</label>
<input type="password" id="confirm-password" name="confirm-password" required>
</div>
<button type="submit" class="btn">Register</button>
</form>
</div>
</body>
</html>
```

Get Methode & Post Method Handler:

```
import java.io.IOException;

import java.io.PrintWriter;

import javax.servlet.ServletException;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;

import javax.servlet.http.HttpServletRequest;

import javax.servlet.http.HttpServletResponse;

@WebServlet("/register")

public class GetPostHandler extends HttpServlet {

@Override

protected void doGet(HttpServletRequest request, HttpServletResponse response)

throws ServletException, IOException {

response.setContentType("text/html");

PrintWriter out = response.getWriter();

out.println("<html><body>");

out.println("<h2>Use the form to register</h2>");

out.println("<a href=\"index.html\">Go to Registration Form</a>");
```

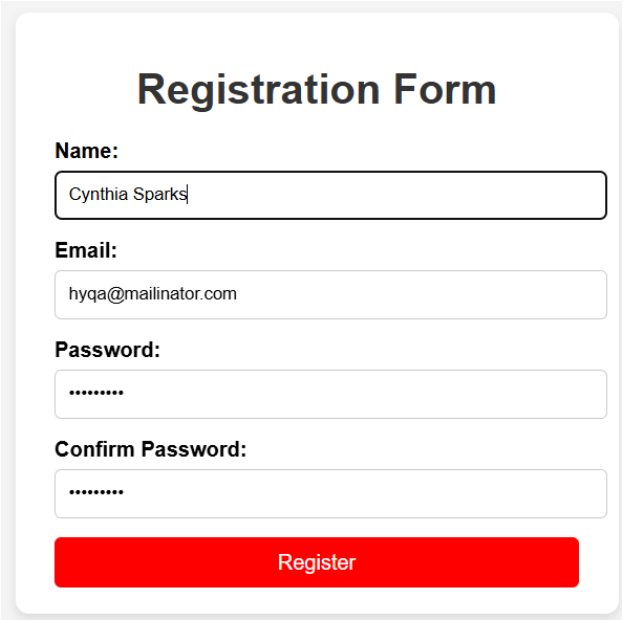
```
out.println("</body></html>");

}

@Override

protected void doPost(HttpServletRequest request, HttpServletResponse response)

throws ServletException, IOException {

response.setContentType("text/html");

PrintWriter out = response.getWriter();

// Get form data

String name = request.getParameter("name");

String email = request.getParameter("email");

String password = request.getParameter("password");

String confirmPassword = request.getParameter("confirm-password");

// Start HTML response

out.println("<html>");

out.println("<head>");

out.println("<title>Registration Result</title>");

out.println("<style>");

out.println("body { font-family: Arial, sans-serif; background-color: #f4f4f4; margin: 0;
padding: 0; display: flex; justify-content: center; align-items: center; height: 100vh; }");
```

```
out.println(".container { background: #fff; padding: 20px; border-radius: 10px; box-shadow: 0
4px 8px rgba(0, 0, 0, 0.1); width: 100%; max-width: 500px; text-align: center; }");

out.println("table { width: 100%; border-collapse: collapse; margin-top: 20px; }");

out.println("th, td { border: 1px solid #ddd; padding: 8px; text-align: left; }");

out.println("th { background-color: #f2f2f2; }");

out.println("h2 { color: #333; }");

out.println("a { display: inline-block; margin-top: 20px; text-decoration: none; color:
#007BFF; }");

out.println("a:hover { color: #0056b3; }");

out.println("</style>");

out.println("</head>");

out.println("<body>");

out.println("<div class='container'>");

// Handle form submission

if (password.equals(confirmPassword)) {

out.println("<h2>Registration Successful!</h2>");

out.println("<table>");

out.println("<tr><th>Field</th><th>Value</th></tr>");

out.println("<tr><td>Name</td><td>" + name + "</td></tr>");
```

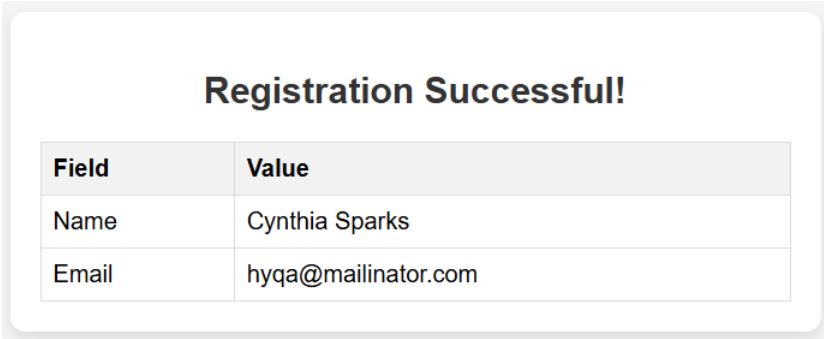
```
out.println("<tr><td>Email</td><td>" + email + "</td></tr>");

out.println("</table>");

} else {

out.println("<h2>Passwords do not match!</h2>");

out.println("<a href=\"index.html\">Go back to the form</a>");

}

// End container and body

out.println("</div>");

out.println("</body>");

out.println("</html>");

}

}
```

4. Result:

Get Method



Post Method



5. Learning Outcomes (what I have learned):

- Understand how to handle GET and POST requests in a Java servlet.
- Learn to retrieve form data from HTTP requests and validate it.
- Gain experience in generating dynamic HTML content using Java's PrintWriter for
- server-side rendering.